

EXHIBIT A

UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.

Before the Honorable Clark S. Cheney
Chief Administrative Law Judge

In the Matter of

CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF

Investigation No. 337-TA-1500

DECLARATION OF DR. SANJAY K. BANERJEE
REGARDING CLAIM CONSTRUCTION

I, Sanjay K. Banerjee, Ph.D., declare as follows:

I. INTRODUCTION AND QUALIFICATIONS

1. I am being compensated for my work on this matter at my standard hourly rate of \$700/hr. My compensation is not dependent on the outcome of this case or the content of my opinions or analysis. I do not hold any financial interest in Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., or Newport Fab LLC (collectively, “Respondents”) or Complainant GlobalFoundries U.S. Inc. (“Complainant”).

2. I have been asked by Respondents to provide my opinions regarding how a person of ordinary skill in the art (“POSA”) would have interpreted certain disputed claim terms in U.S. Patent Nos. 9,865,546 (“’546 patent”) and 10,707,167 (“’167 patent”), which are asserted in the above-captioned investigation. I understand that Complainant has asserted infringement by Respondents of all claims (1-6) of the ’167 patent and all claims (1-14) of the ’546 patent.

3. I reserve the right to supplement or amend this declaration to address any evidence, opinion, statement, or position taken by Complainant or any expert who submits a declaration in support of Complainant's claim construction positions.

4. I have considered and relied upon my education, background, and experience in forming the opinions expressed in this declaration. In addition, I have reviewed and relied upon the following materials:

- the asserted patents and their prosecution histories;
- the claim terms and proposed constructions of Respondents and Complainant; and
- any other evidence discussed or cited in this declaration, including the evidence listed in Section VII, below.

5. I currently serve as the Cockrell Family Regents Chair Professor of Electrical and Computer Engineering. I was also the Director of the Microelectronics Research Center at the University of Texas at Austin ("UT") from September 1999 through January 2025. Prior to this position, I was first an Assistant Professor (September 1987 – August 1990) and later an Associate Professor (September 1990 – August 1993) at UT, before being named a Professor in September 1993. Prior to starting my academic work at UT, I was a member of the technical staff at Texas Instruments (Corporate R&D) from the time I completed my doctoral studies in 1983 until August 1987. I worked on vertical 3D Dynamic Random Access Memory (DRAM) cells (*see* Ex. 35, S. Banerjee & M. Bordelon, *A Model for the Trench Transistor*, 34 IEEE Trans. Elec. Dev. 2485, 2485-2493 (December 1987)).

6. In addition to my academic work at UT, I was also the Director of the Southwest Academy of Nanoelectronics, one of three centers created by the Semiconductor Research Corporation Nanoelectronics Research Initiative in 2006 to research and develop a replacement

for conventional metal oxide semiconductor field effect transistors (“MOSFETs”) during the entire program.

7. I received my Ph.D. in Electrical Engineering from the University of Illinois in 1983. I also received an M.S. in Electrical Engineering from the University of Illinois in 1981. I received a B. Tech. in Electronics from the Indian Institute of Technology at Kharagpur in 1979.

8. As described in my curriculum vitae (“CV”) (Ex. D), I have more than 40 years of experience in the field of electrical engineering, including extensive experience in semiconductor development and fabrication including that of 3-Dimensional Dynamic Random Access Memory (“DRAM”) cells, MOSFETs, flash memory, and beyond-complementary metal-oxide-semiconductor (“CMOS”) transistors. My work in these fields has resulted in thirty-five United States patents related to methods of forming polysilicon and novel transistors, flash memory, and vertical DRAM cells. I have been elected a Fellow of the National Academy of Inventors.

9. Much of my research and experience has related to methods for MOSFET fabrication and design, including deposition of thin films by chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), patterning using photolithography and etching using plasma-based Reactive Ion Etching (RIE) which are some of the processing techniques used to make 3D NAND flash. I have worked extensively on polysilicon thin film transistors as well as high-k and other dielectrics used for charge storage and tunneling. A number of the sponsored research grants on which I have been a principal or coinvestigator have related to chemical deposition methods and/or ultra-shallow source/drain junction technology, including “Transport in MOSFETs” (Motorola, August 1993 – August 1994), “Ultra Shallow Junction Technology” (SEMATECH January 1994 – December 1996), and “Vertical Si-Ge-C MOSFETS” (Semiconductor Research Corporation September 2000 – August 2003).

10. I am a Fellow of the Institute of Electrical and Electronics Engineers (“IEEE”), the American Physical Society (“APS”), and the American Association for the Advancement of Science (“AAAS”). I have served on numerous professional and government committees in the electrical engineering field, including chairing multiple IEEE committees, meetings, and programs, serving on Morgan & Claypool’s editorial board, and serving as a member on the International Technology Roadmap for Semiconductors (“ITRS”).

11. I have taught courses in Semiconductor Physics, Solid State Electronic Devices, and Microelectronic and VLSI Device Fabrication at UT. In addition, I have delivered many courses to industry on Semiconductor Devices and Memory and Semiconductor Processing. I have taught courses that cover 3D memories, such as 3D NAND (Ultra Large Scale Integration Techniques and Solid State Electronic Devices). As a principal investigator, I have been the advisor to over 80 Doctoral and Masters students.

12. I am the co-author of several books and invited book chapters covering the area of electronic and semiconductor devices, including: Dopant Diffusion, Solid State Electronic Devices (three editions), Silicon-germanium Devices, and Novel 3D CMOS. I have authored or co-authored more than 1,000 papers and presentations in the areas of semiconductor devices and electronics development over the course of my career.

13. I have been the recipient of fifty grants to fund my research from a variety of funding agencies including the National Science Foundation (“NSF”), DARPA, the Department of Energy, Semiconductor Research Corporation, the Department of Defense-Multidisciplinary University Research Initiative, and the Office of Naval Research.

14. I have received numerous awards for my work, including the NSF Presidential Young Investigator Award. The NSF Presidential Young Investigator Award was a highly

competitive honor bestowed annually upon young academic leaders in science and engineering. I have received numerous other awards, including some of the highest honors for semiconductor devices such as the Andrew Grove Award from the IEEE, and the Semiconductor Industry Association/Semiconductor Research Corporation University Researcher Award, for my work in semiconductor memory and transistors.

15. My education, employment history, publications, patents, awards, and other relevant information are further detailed in my curriculum vitae (“CV”), a copy of which is provided as Exhibit D to this declaration and which I hereby incorporate by reference.

II. APPLICABLE LEGAL PRINCIPLES

16. I am not a lawyer and am not offering legal opinions. However, I was asked by counsel for Respondents to provide opinions in the context of certain legal standards summarized below. I rely upon these legal standards as background for my analysis and to assist in forming my opinions regarding claim construction. Counsel has informed me that claim construction is a question of law for the Chief Administrative Law Judge (“CALJ”) to decide in this investigation, with potential underlying issues of fact. I understand from counsel that the purpose of claim construction is to determine the meaning and scope of the asserted claims from the perspective of a POSA.

17. I have been informed by counsel that a disputed claim term should be read not only in the context of the specific claim in which it appears, but also in the context of the entire patent-in-suit, including the specification. I understand from counsel that the specification has been referred to in some cases as the single best guide to the meaning of a disputed term. As such, I understand from counsel that, for terms without customary meaning in the art, the specification typically supplies the best context for understanding claim meaning. I have been informed by

counsel that ALJs ordinarily interpret claim terms consistently across patents having the same specification. The patents addressed in this declaration share a substantively identical specification.

18. I understand from counsel that patents' file histories can additionally inform the meaning of claim language by demonstrating how the inventors understood the alleged inventions and whether the inventors limited the inventions during the course of prosecution, making claim scope narrower than it otherwise would be. I also understand from counsel that the inventor may expressly or implicitly define a term in the specification of a patent and that this is generally known as "lexicography." I have been informed by counsel that, when the meaning of a claim term is not clear from the intrinsic record, the CALJ may also consult extrinsic evidence, such as this declaration, in construing the term.

19. I understand from counsel that claim construction must be understood from the perspective of a POSA at the time of alleged invention. I also understand from counsel that in determining who would have been a POSA, it is proper for me to consider the type of problems encountered in the art, prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the education level of active workers in the field. I have provided my opinion below as to who I would have considered to be a POSA in the field of the asserted patents.

20. Finally, I understand from counsel that a patent claim is invalid as indefinite if the claim, as read in light of the specification delineating the patent and the prosecution history, fails to inform, with reasonable certainty, those skilled in the art about the scope of the invention. I have been informed by counsel that a claim term does not become reasonably certain simply because a skilled artisan, when pressed, might manage to articulate a definition for it. I understand

from counsel that showing that a skilled artisan cannot recognize the scope of a claim term with reasonable certainty requires showing considerably less uncertainty than showing that a skilled artisan could not understand the claim term at all.

III. PERSON OF ORDINARY SKILL IN THE ART

21. As noted above, I understand from counsel that the asserted claims must be construed from the point of view of a POSA as of the earliest claimed priority date of the '546 and '167 patents, which I have assumed to be June 3, 2015 (the filing date of the application that issued as the '546 patent, to which the application that issued as the '167 patent claims priority). I have been informed by counsel that Complainant does not allege that the '546 or '167 patents are entitled to a priority date earlier than June 3, 2015.

22. The relevant technical field for the asserted patents is the design and fabrication of semiconductor devices, particularly the formation of contacts in semiconductor devices. In my opinion, based on the foregoing legal principles, my experience and knowledge of the field and those who work and study in it, and the materials considered in preparing this declaration, a POSA in this technical field on June 3, 2015, would have had at least a Bachelor of Science in electrical engineering, materials science, materials engineering, materials science and engineering, chemical engineering, or similar majors with 2-3 years of experience related to the design and fabrication of semiconductor devices, including contact formation. More education could substitute for experience or vice versa.

23. My opinions in this declaration would not change if the POSA were determined to have somewhat more or less or slightly different education and/or experience than what I have summarized above. My knowledge and skill exceed that of a POSA. But my opinions herein reflect and are made from the perspective of a POSA as of June 3, 2015.

IV. SUMMARY OF THE '546 AND '167 PATENTS' COMMON SPECIFICATION

24. I have reviewed the common specification of the '546 and '167 patents and summarize herein portions of it that are relevant to the disputed claim terms of these patents. The '546 and '167 patents relate to “contacts over a semiconductor substrate and methods of forming the same.” '546 patent, 1:8-9. “More specifically, the present invention relates to forming an intermix region at an interface of a bottom portion of contacts and an upper portion of the semiconductor substrate, respectively.” *Id.*, 1:9-13.

25. The patents acknowledge use of “Ti-based silicide liner” in the prior art to form contacts in contact openings created over prefabricated source and drain regions of transistors. *Id.*, 1:30-34. The patents also acknowledge the prior art’s use of “a bilayer of NiPt and Ti” “to coat contact openings over the source and drain” as an improvement for 14/10 nm silicon-on-insulator PFET applications. *Id.*, 1:35-40. However, according to the patents, “this bilayer requires deposition of a relatively thick metal layer.” *Id.*, 1:40-41. The patents continue: “As semiconductor devices get smaller, requiring a thick bilayer in the contact causes contact opening fill issues, such as an incomplete contact fill.” *Id.*, 1:41-44.

26. To solve this problem, according to the patents, “[a]spects of the present invention utilize an intermix concept as a template for formation of a contact to a semiconductor substrate of a semiconductor structure.” *Id.*, 2:35-37. More specifically, the claimed inventions relate to “depositing a first liner layer in a contact opening such that an intermix region is formed at an interface of the bottom portion of the contact opening and an upper portion of the semiconductor substrate thereunder, e.g., source and drain.” *Id.*, 2:44-48. “Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact.” *Id.*, 2:48-50. “Therefore, aspects of the present invention do not increase the deposited metal thickness in the

contact openings.” *Id.*, 2:50-52. Thus, according to the patents, “contact opening fill issues are avoided and overall contact resistance is improved.” *Id.*, 2:52-53.

27. The patents refer to various “layers” used to make the devices recited in the claims of the ’546 and ’167 patents. For example, Figure 1 depicts a “dielectric layer” as item 120 “over substrate 112.” *Id.*, 2:56-57.

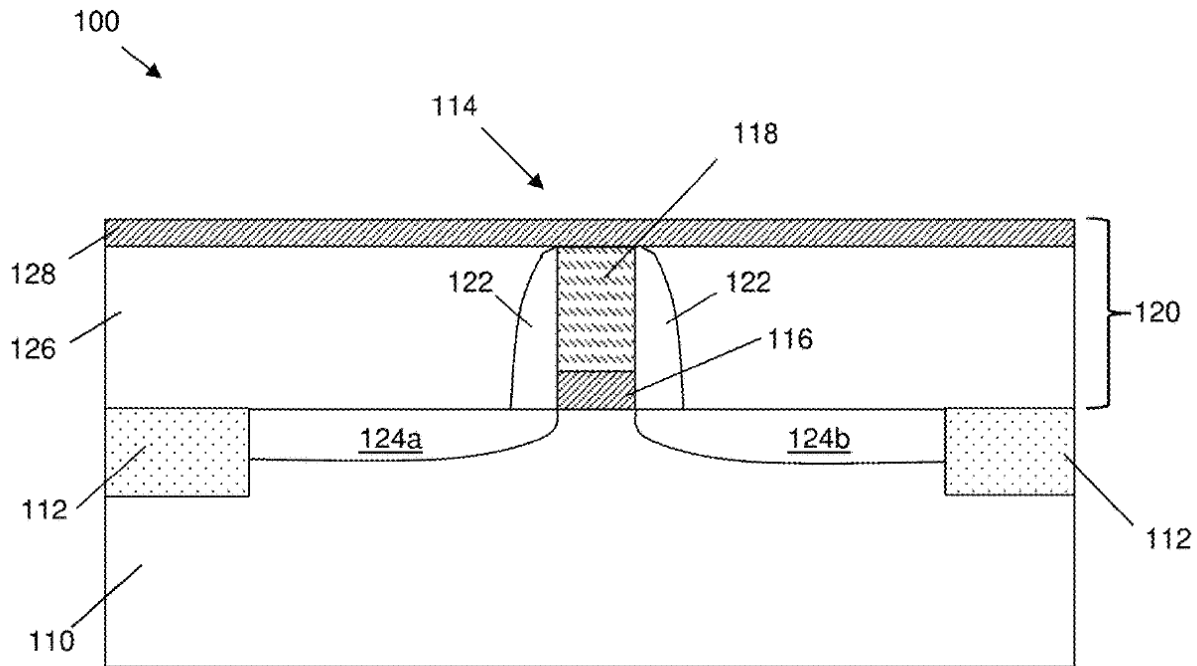


FIG. 1

“In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128.” *Id.*, 4:56-58. “While shown as two separate layers, it is to be understood that dielectric layer 120 is not limited to only two layers.” *Id.*, 4:58-60. “Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers.” *Id.*, 4:60-62. In other words, the term “layer,” as used in the patents, could mean a “single layer,” two layers, or “more than two” layers. *Id.*, 4:56-62. There is no explanation in the patents about how to distinguish between different “layers”

within a larger “layer.” For example, layers within a larger layer could comprise the same material, making it impossible for a POSA to distinguish between layers when attempting to assess infringement or non-infringement.

28. Contact openings (items 130a and 130b in Figure 2 below) may be formed “by lithography and etching” through dielectric layer (120) to expose source and drain regions (124a and 124b). *Id.*, 5:39-40.

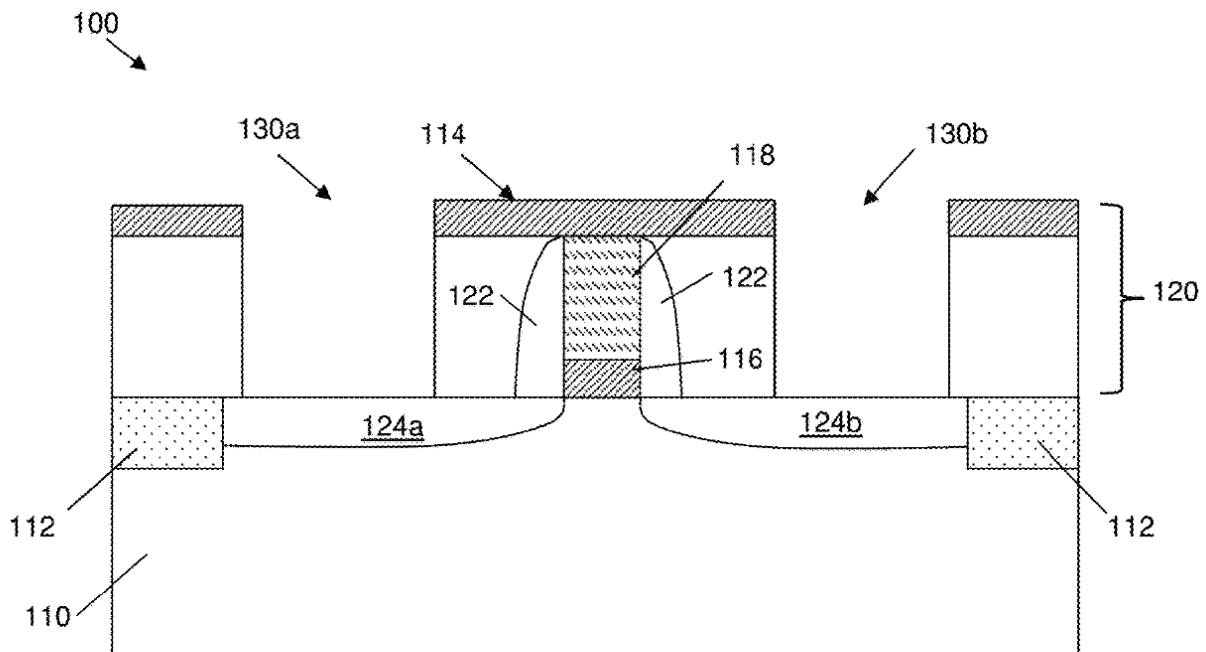


FIG. 2

29. Thereafter, “a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128,” as shown in Figure 3. *Id.*, 5:57-59.

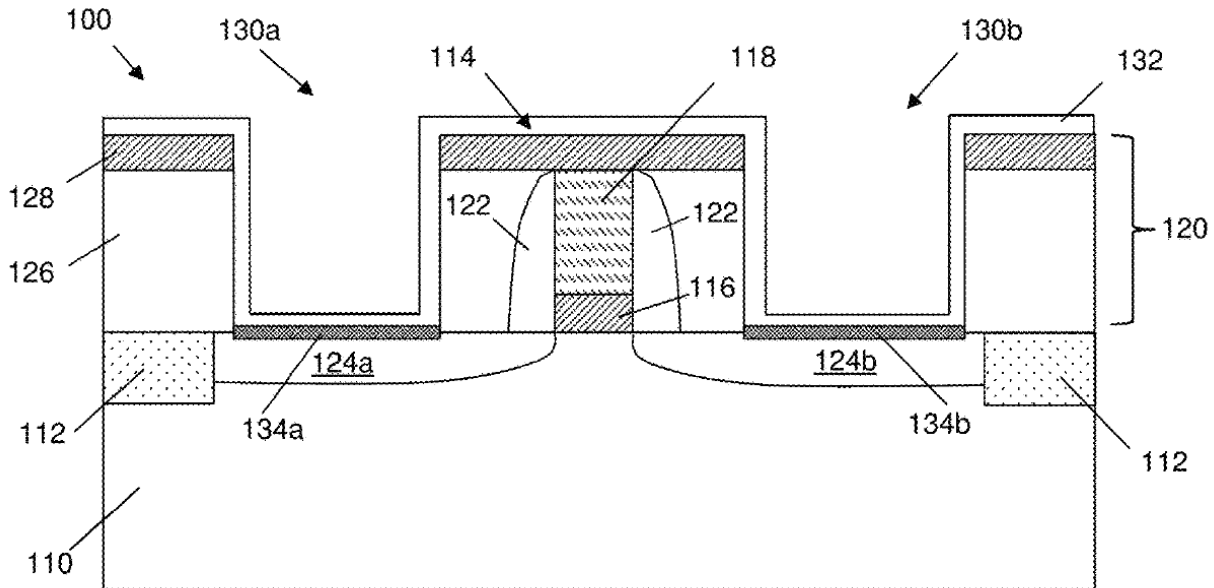


FIG. 3

“First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb.” *Id.*, 5:64-65. Again, the first liner layer could comprise one or more layers of any of these materials or any other material(s). There is no requirement for the different sublayers of a “first liner layer” to be distinguishable or comprised of different materials.

30. According to the patents: “First liner layer 132 may be deposited such that a first intermix region 134a, 134b is formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.” *Id.*, 5:65-6:2. “First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b.” *Id.*, 6:3-6. “Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder.” *Id.*, 6:6-9. “In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for

the respective source and drain 124a, 124b thereunder.” *Id.*, 6:9-13. “While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent.” *Id.*, 6:13-15. “The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness, for example.” *Id.*, 6:15-19.

31. The common specification continues: “In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition).” *Id.*, 6:19-21. The patents put no upper limit on the amount of “Si content” or substrate material an “intermix region” may contain while still qualifying as an “intermix region.” The patents state: “In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.).” *Id.*, 6:21-28. “In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity.” *Id.*, 6:28-31. “First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystallize intermix region 134a, 134b into a lower resistance layer as will be described herein.” *Id.*, 6:31-34. In sum, the common specification puts no meaningful limits on what could qualify as an “intermix region.” It could be composed of any materials and be created via any method.

32. The specification further states: “First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard sputtering tools.” *Id.*, 6:35-37. However, the specification does not explain what the inventors

considered to be “standard sputtering tools” and what tools were not “standard.” *Id.* The specification continues: “First intermix regions 134a, 134b are much more conformal on Si structures that are not planar.” *Id.*, 6:37-39. “Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices.” *Id.*, 6:39-43. “Encroachment of silicide under spacers 122 and encroachment on defective silicon areas may be substantially reduced or eliminated by this intermixing concept, thus increasing the yield of semiconductor structures or devices.” *Id.*, 6:43-47.

33. The specification confusingly refers to both the originally deposited “first liner layer” and the undiffused “first liner layer” as being a “first liner layer,” even though the latter does not include portions of the original first liner layer that have diffused into substrate. *Id.*, 6:48-63. The specification states: “Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown.” *Id.*, 6:48-49. Figure 4 is depicted below:

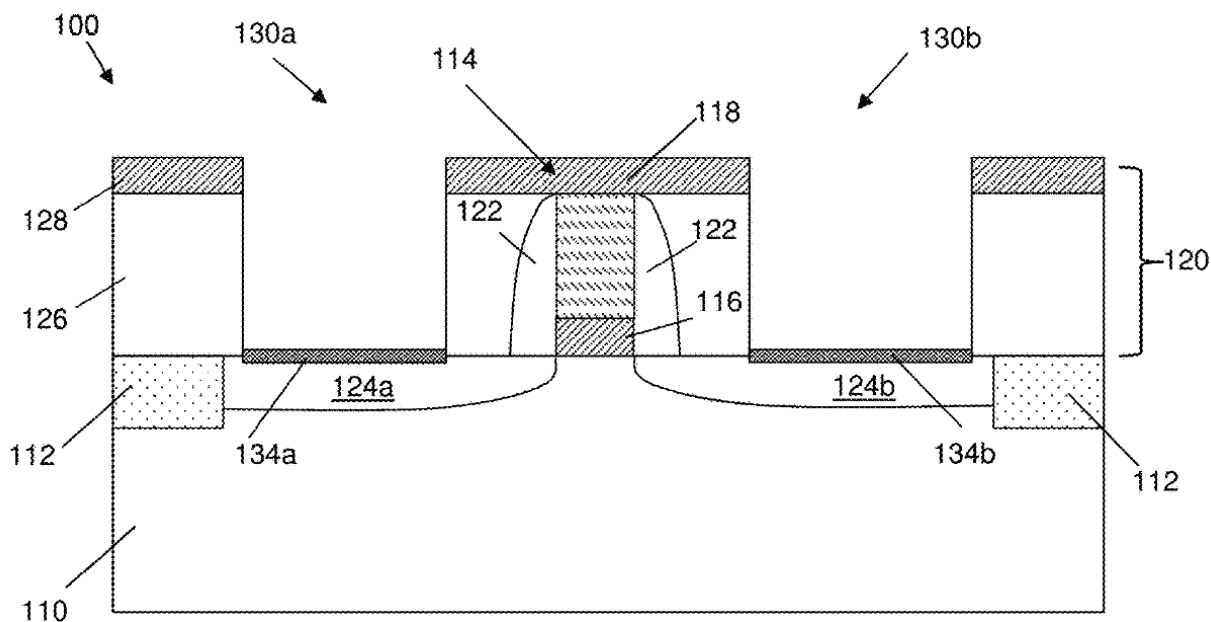


FIG. 4

The specification states: “In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128.” *Id.*, 6:49-52.

34. The specification further describes deposition of a “refractory metal layer.” *Id.*, 6:66. “As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128 and first intermix regions 134a, 134b at the bottom portion of each contact opening 130a, 130b.” *Id.*, 6:64-7:3.

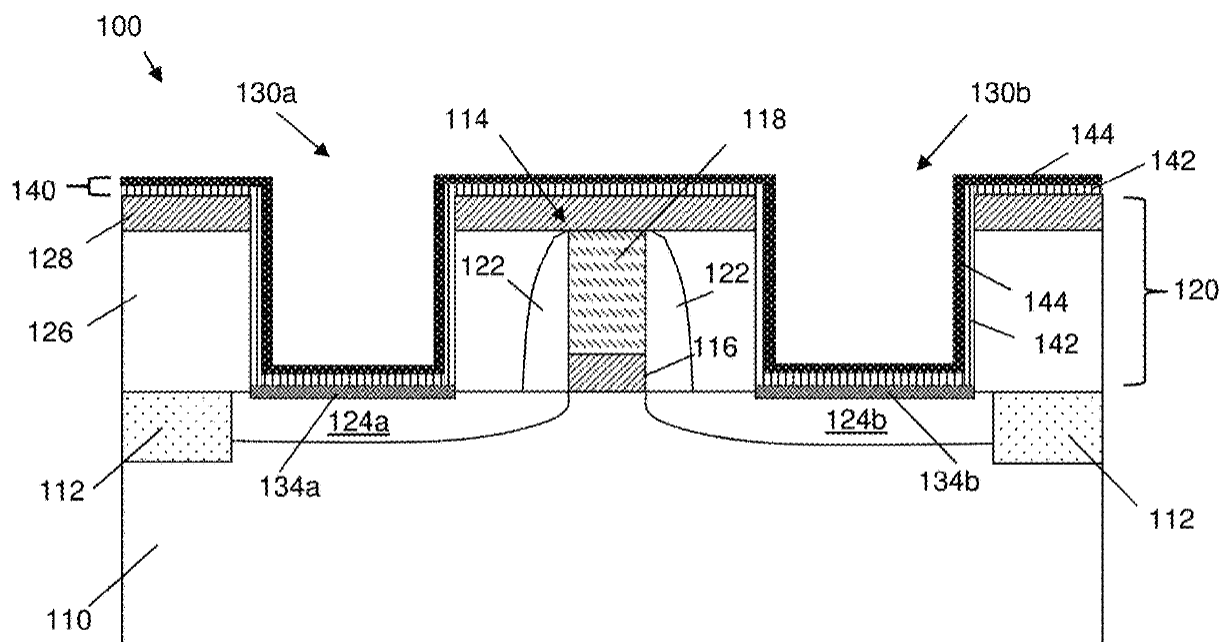


FIG. 5

According to the specification, “[r]efractory metal layer 140 may include, but is not limited to: W, Ti, Ta or Mo employed directly as a ‘metal’; a refractory metal employed as a constituent of a ‘compound’ thereof, e.g., TiN, TaN, etc.; or even as a constituent of an ‘alloy’ with another metal, e.g., titanium-tungsten.” *Id.*, 6:64-7:3. In other words, the refractory metal layer can be composed

of any metal.¹ “In some embodiments, as shown in FIG. 5, refractory metal layer 140 may include a bilayer of a first refractory material 142 and a second refractory material 144 such that first refractory metal material 142 substantially separates first liner layer 132 from second refractory metal layer 144.” ’546 patent, 7:3-8. “For example, in some embodiments, first refractory metal material 142 may include Ti and second refractory metal material 144 may include TiN.” *Id.*, 7:15-17. “While shown as a bilayer, it is to be understood that refractory metal layer 140 may include any number of layers of refractory metal materials without departing from aspects of the invention.” *Id.*, 7:20-23. This portion of the specification comports with the other uses of “layer” in that “layer” can mean “any number of layers of refractory metal materials” (which, as noted above, according to the specification, could comprise any metals). There is no requirement that multiple sublayers in a “refractory metal layer” comprise different metals or that the sublayers be distinguishable from one another, making it impossible for the POSA to assess infringement or non-infringement.

35. The specification continues: “As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region 154a, 154b is formed at the interface of the bottom portion of contacts 170a, 170b and upper portion of source 124a and drain 124b.” *Id.*, 7:59-63.

¹ The plain and ordinary meaning of “refractory metal” is different from the meaning attributed to this phrase in the ’546 and ’167 patents. A POSA would have understood “refractory metal” to refer to “metals with melting points around 2000 °C and higher, including Tungsten, Molybdenum, Chromium, Tantalum, Niobium, and Rhenium, along with their refractory alloys.” Ex. 2, *Refractory Metal*, ScienceDirect, <https://www.sciencedirect.com/topics/materials-science/refractory-metal> (last visited August 24, 2026). But the ’546 and ’167 patents use a different definition of “refractory metal” that includes titanium, which does not have a melting point of around 2000 °C or higher. ’546 patent, 7:15-17; Ex. 3, *Titanium*, Royal Society of Chemistry, <https://periodic-table.rsc.org/element/22/titanium> (last visited August 24, 2026). This is further evidence that a POSA would not have understood the term “refractory metal layer” to have a plain and ordinary meaning in light of the ’546 and ’167 patents. To the contrary, it appears to be a coined term with no reasonably certain meaning.

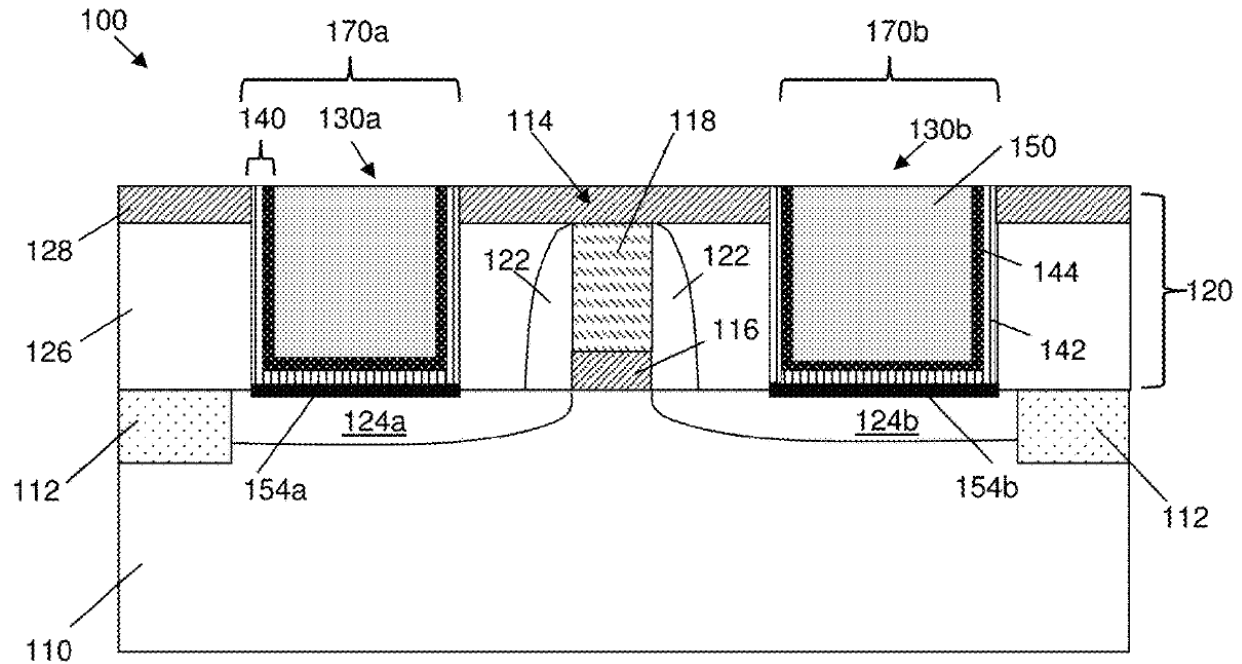


FIG. 8

“The laser anneal process may transform the composition of first intermix region 134a, 134b (FIGS. 3-7) into a second composition resulting in second intermix region 154a, 154b in place of first intermix region 134a, 134b.” *Id.*, 7:64-8:1. “That is, the laser anneal process may crystallize intermix regions 134a, 134b into a lower resistance layer, i.e. a second intermix region 154a, 154b.” *Id.*, 8:1-3. “Second intermix regions 154a, 154b may have compositions including a mixture of the materials used for source 124a and drain 124b, first liner layer 132, and refractory metal layer 140.” *Id.*, 8:3-6. “In an embodiment where Ti is used as refractory metal layer 140, second intermix region 154a, 154b may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi.” *Id.*, 8:6-9. “That is, during the laser anneal process, refractory metal 140 may diffuse into first intermix regions 134a, 134b (FIGS. 3-7) to create second intermix regions 154a, 154b.” *Id.*, 8:9-12.

36. The specification includes an embodiment in which “first intermix region 134a, 134b may include TiSi and second intermix region 154a, 154b may include TiSi with a higher concentration of Ti.” *Id.*, 9:1-4. The specification does not require the first intermix region to have a composition that is different from the second intermix region. The specification provides no explanation as to how much or what type of intermixing is required for something to qualify as an “intermix region.” There is no requirement regarding the homogeneity (or lack thereof) of any “intermix region” in terms of how uniformly components of the region are distributed.

V. SUMMARY OF THE ’546 AND ’167 PATENTS’ PROSECUTION HISTORIES

37. I have briefly summarized the file histories of the ’546 and ’167 patents below.

A. Prosecution History of the ’546 Patent

38. The application that issued as the ’546 patent was filed on June 3, 2015. On April 21, 2016, the US Patent Office issued a Requirement for Restriction/Election, given that Complainant sought claims on both semiconductor structures and methods for forming contacts to semiconductor substrate. Ex. 13, GF-ITC-00004865 – 68. On June 20, 2016, Complainant chose to pursue method claims 1-14. *Id.* at GF-ITC-00004871 – 72.

39. On September 26, 2016, the examiner rejected all pending claims as being anticipated and/or obvious in view of Ex. 4, U.S. Patent Application Publication No. 2012/0161324 A1 (“Heinrich”) alone or in combination with Ex. 5, U.S. Patent Application Publication No. 2014/0027883 A1 (“Hu”). Ex. 13, GF-ITC-00004878 – 84.

40. On December 8, 2016, Complainant responded to the examiner’s rejection of September 26, 2016 by making non-substantive, typographical edits to the claims and arguing, among other things, that the primary reference relied upon by the examiner (Heinrich), as mapped

to the pending claims by the examiner, did not teach the use of a first liner layer to create an intermix region. *Id.* at GF-ITC-00004908 – 13.

41. On May 8, 2017, the examiner issued a final rejection relying upon the same prior art references and largely the same arguments. *Id.* at GF-ITC-00004925 – 33.

42. After an interview with Complainant’s representative, the examiner issued a Notice of Allowance on October 19, 2017, asserting that Heinrich “failed to teach depositing a refractory metal layer over the first intermix region of claim 1.” *Id.* at GF-ITC-00004978. The ’546 patent issued on January 9, 2018.

B. Prosecution History of the ’167 Patent

43. The application that issued as the ’167 patent was filed as a continuation of the ’546 patent’s application on November 30, 2017. On August 27, 2018, the examiner rejected pending claims 1-6 of the ’167 patent as being anticipated by Ex. 6, U.S. Patent Application Publication No. 2015/0221740 A1 (“Alptekin”). Ex. 14, GF-ITC-00004210 – 11.

44. In response, Complainant amended sole independent claim 1 to recite: “wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.” *Id.* at GF-ITC-00004243. Complainant then argued, among other things, that Alptekin did not disclose this new claim element. *Id.* at GF-ITC-00004245 – 47.

45. On March 23, 2020, the examiner issued a Notice of Allowance, stating that “the prior arts of record do not teach or fairly suggest a limitation of: wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal

material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.” *Id.* at GF-ITC-00004256 – 57. The examiner also stated: “There is no reason apparent to the examiner that all the features claimed in claim 1 would be obvious over the prior art,” after quoting a portion of the common specification reciting: “Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.” *Id.* at GF-ITC-00004257 (quoting ’546 patent, 2:48-53). The ’167 patent issued on July 7, 2020.

VI. ASSERTED CLAIMS OF THE ’546 AND ’167 PATENTS

46. As noted above, I have been informed by counsel that Complainant is asserting claims 1-14 of the ’546 patent and claims 1-6 of the ’167 patent. I list the asserted claims below.

A. Asserted Claims of the ’546 Patent

47. Asserted claims 1-14 of the ’546 patent recite as follows:

1. A method for forming a contact in a dielectric layer to a semiconductor substrate, the method comprising:

forming a contact opening in the dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate;

depositing a first liner layer to conformally coat the contact opening

causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate;

depositing a refractory metal layer over the first intermix region; and

depositing a metal in the contact opening thereby forming the contact.

2. The method of claim 1, further comprising: removing a remaining non-diffused portion of the first liner layer to expose the first intermix region at the upper portion of the semiconductor substrate prior to depositing the refractory metal layer.

3. The method of claim 1, wherein the depositing the first liner layer includes depositing at least one of: nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

4. The method of claim 1, wherein depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

5. The method of claim 1, wherein depositing the first liner layer includes performing physical vapor deposition.

6. The method of claim 1, further comprising:

polishing the metal; and

performing a laser anneal of the opening such that a second intermix region is formed at the upper portion of the semiconductor substrate and a bottom surface of the contact, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

7. The method of claim 1, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

8. A method for forming a set of contacts in a dielectric layer to a field effect transistor, the field effect transistor including a source, a drain, and a gate between the source and the drain, the method comprising:

forming a contact opening in the dielectric layer to expose an upper-portion of each of the source and the drain;

depositing a first liner layer to conformally coat each contact opening and causing a portion of the first liner layer to diffuse into the upper portion of each of the source and the drain to form a first intermix region at the upper portion of the each of the source and the drain;

depositing a refractory metal layer over the first intermix region; and

depositing a metal to substantially coat the refractory metal layer and substantially fill each of the openings thereby forming the set of contacts.

9. The method of claim 8, further comprising: removing a remaining non-diffused portion of the first liner layer to expose the intermix region prior to depositing the refractory metal layer.

10. The method of claim 8, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), nickel (Ni), titanium (Ti), titanium silicide (TiSi), cobalt (Co), erbium (Er), platinum (Pt), and ytterbium (Yb).

11. The method of claim 8, wherein the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN).

12. The method of claim 8, wherein the depositing the first liner layer includes performing physical vapor deposition.

13. The method of claim 8, further comprising:

polishing the metal in each of the openings; and

performing a laser anneal of each of the openings such that a second intermix region is formed at the upper portion of the respective source and drain thereunder, wherein the second intermix region is composed of materials of the first intermix region and a material used for the refractory metal layer.

14. The method of claim 13, wherein the depositing the first liner layer includes depositing nickel platinum (NiPt), the depositing the refractory metal layer includes depositing a bilayer of titanium (Ti) and titanium nitride (TiN), and the depositing the metal in the opening includes depositing tungsten (W).

B. Asserted Claims of the '167 Patent

48. Asserted claims 1-6 of the '167 patent recite as follows:

1. A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

a contact opening in a dielectric layer over the semiconductor substrate;

a first liner layer substantially coating a sidewall of the contact opening;

a refractory metal layer substantially coating the first liner layer in the contact opening;

a metal substantially filling the contact opening; and

an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;

wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

2. The contact of claim 1, wherein the intermix region includes a titanium-based silicide.
3. The contact of claim 1, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).
4. The contact of claim 1, wherein the first liner layer includes nickel platinum (NiPt), the first refractory metal material includes titanium (Ti) and the second refractory metal material includes titanium nitride (TiN).
5. The contact of claim 1, wherein the metal includes tungsten (W).
6. The contact of claim 1, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

VII. MATERIALS CONSIDERED

49. In arriving at my opinions in this declaration, I have considered the asserted patents, their file histories, the parties' proposed claim constructions, and all other information and documents cited or discussed in this declaration, including the exhibits below.

Exhibit No.	Description
N/A	Complaint, <i>Certain Semiconductor Devices, Products Containing Same, and Components Thereof</i> , Investigation No. 337-TA-1500 ("Complaint"), and accompanying exhibits and appendices
N/A	Certified copy of U.S. Patent No. 9,865,546 (Exhibit 4 to Complaint) ("546 patent")
N/A	Certified copy of U.S. Patent No. 10,707,167 (Exhibit 6 to Complaint) ("167 patent")

N/A	Certified copy of Prosecution History of the U.S. Patent No. 9,865,546 (Appendix D to Compl.) (GF-ITC-00004771 – GF-ITC-00005023)
N/A	Certified copy of Prosecution History of the U.S. Patent No. 10,707,167 (Appendix F to Compl.) (GF-ITC-00004128 – GF-ITC-00004309)
D	Dr. Sanjay K. Banerjee Curriculum Vitae
1	Complainant's Infringement Contentions, Ex. A4
2	https://www.sciencedirect.com/topics/materials-science/refractory-metal
3	https://periodic-table.rsc.org/element/22/titanium
4	U.S. Patent Application Publication No. 2012/0161324 A1 (“Heinrich”)
5	U.S. Patent Application Publication No. 2014/0027883 A1 (“Hu”)
6	U.S. Patent Application Publication No. 2015/0221740 A1 (“Alptekin”)
7	<i>Intellectual Ventures I LLC, v. Canon Inc.</i> , C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. at 9-10 (D. Del. Apr. 10, 2014), Dkt. No. 252
8	<i>BridgeLux, Inc., v. Cree, Inc.</i> , C.A. No. 06-6495-PJH, Ord. Construing Claims at 10 (N.D. Cal. Aug. 15, 2008), Dkt. No. 109
9	<i>Advanced Technology Incubator, Inc. v. Sharp Corporation</i> , C.A. No. 2:7-cv-468 (E.D. Tex.)
12	Merriam-Webster's Collegiate Dictionary (11th Ed.), 2003 (definition of “intermix”)
13	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 9,865,546 (GF-ITC-00004771 – GF-ITC-00005023)
14	<i>Excerpt</i> of Prosecution History of the U.S. Patent No. 10,707,167 (GF-ITC-00004128 – GF-ITC-00004309)
15	Chow, et al., <i>Silicide Formation from Laser Thermal Processing of Ti/Co Bilayers</i> (2004)
34	Ashwin Shah et al., <i>A 4Mb DRAM with cross point trench transistor cell</i> , SC-21:5 IEEE J. ISSCC Dig. Tech. Papers, 618-626 (1986)
35	Shaw Banerjee & D. Mark Bordelon, <i>A Model for the Trench Transistor</i> , ED-34:12 IEEE Transactions on Electron Devices 2485, 2485-2492 (1987)
36	Shaw Banerjee et al, <i>Leakage Mechanisms in the Trench Transistor DRAM Cell</i> , ED-35:1 IEEE Transactions on Electron Devices, 108-116 (1988)

VIII. CLAIM TERMS AND PROPOSED CONSTRUCTIONS

50. I understand that the parties have proposed the following claim constructions for the following disputed terms in the '546 and '167 patents:

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“layer” / “first liner layer” / “refractory metal layer” ('546 patent, claims 1-14; '167 patent, claims 1, 3, and 4)	Plain and ordinary meaning; not indefinite.	Indefinite Alternatively, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”
“intermix region” ('546 patent, claims 1, 2, 6, 8, 9, and 13; '167 patent, claims 1, 2, and 6)	Plain and ordinary meaning; not indefinite.	Indefinite

51. I have been informed by counsel that the parties have not agreed upon a construction for any claim term in the '546 and '167 patents. Below, I provide opinions about the parties' proposed constructions for the disputed claim terms above from the '546 and '167 patents. I understand from counsel that Respondents dispute indefiniteness and claim construction for other terms in the '546 and '167 patents but that, to narrow disputes, only the above two terms have been identified for claim construction briefing at this time. I reserve the right to address other disputed issues of indefiniteness and claim construction if I am asked and permitted to do so.

IX. CLAIM CONSTRUCTION

A. “layer” / “first liner layer” / “refractory metal layer”

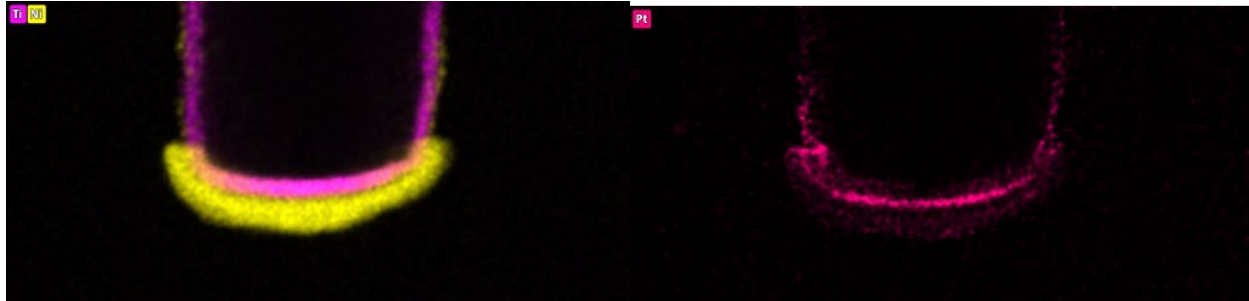
Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“layer” / “first liner layer” / “refractory metal layer” ('546 patent, claims 1-14; '167 patent, claims 1, 3, and 4)	Plain and ordinary meaning; not indefinite.	Indefinite or, if not indefinite: “a separate and defined [first liner / refractory metal] coating of material”

52. In my opinion, the terms “layer,” “first liner layer,” and “refractory metal layer” should be construed in accordance with their plain and ordinary meanings: “a separate and defined [first liner / refractory metal] coating of material.” There is no way by which a POSA would be able to understand these terms if they are not interpreted to have this meaning. If they are not construed in accordance with these plain and ordinary meanings, the terms are indefinite because they fail to inform a POSA about the scope of the alleged inventions with reasonable certainty, when read in light of the patents’ specifications and prosecution histories.

53. The term “layer” has a plain and ordinary meaning. For example, one dictionary defines “layer” to mean “[a] single thickness, coating, or stratum spread out or covering a surface.” Ex. 7, *Intellectual Ventures I LLC, v. Canon Inc.*, C.A. No. 11-792-SLR, Memo. Ord. re Claim Constr. at 9-10 (D. Del. Apr. 10, 2014), Dkt. No. 252 (citing Webster’s Dictionary at 623). Another dictionary defines “layer” as “[a] defined thickness which is part of a material, or which surrounds it.” Ex. 8, *Bridgelux, Inc., v. Cree, Inc.*, C.A. No. 06-6495-PJH, Order Construing Claims at 10 (N.D. Cal. Aug. 15, 2008), Dkt. No. 109 (citing Wiley Electrical and Electronics Engineering Dictionary at 413 (John Wiley & Sons, 2004)). Although the intrinsic evidence is unclear, at best, about whether and when it applies something close to this meaning of “layer,” Complainant’s Infringement and Domestic Industry Contentions interpret these terms in a way that lacks reasonable certainty with respect to the scope of the claims and raise the need for express constructions of these terms.

54. For example, one of Respondents’ accused processes uses nickel, platinum, and titanium at different stages of fabrication. See Ex. 1, Complainant’s Infringement Contentions, Ex. A4 at 4-7. Rather than identifying a single one of these materials as the “first liner layer,” Complainant’s Infringement Contentions arbitrarily identify ***all three*** for the “first liner layer”

claim element, including the unidentified and unexplained screenshot below in their Contentions (“Ti” representing alleged titanium, “Ni” representing alleged nickel, and “Pt” representing alleged platinum). *Id.* at 5.



Complainant has no idea which of these, if any, could be a “first liner layer” as used in the ’546 and ’167 patents because this term has no meaning in the patents according to Complainant. The first liner layer does not even have to be metal, according to the patents. It could be any thickness of any one or more materials that overlap or coincide with any other materials. It would have been impossible for a POSA to determine which materials should be considered a “first liner layer” under Complainant’s interpretation.

55. Arbitrarily combining the first coating of material with *other, subsequent* coatings of materials and labeling the collective coatings as a “first liner layer” (as in Complainant’s Infringement Contentions) reflect an interpretation under which it would be impossible for the POSA to assess what needs to be diffused into the upper portion of the semiconductor substrate to form a “first intermix region.” *See* ’546 patent, claim 1. In addition, if materials *other than* the very first material deposited over a contact opening could be included in the alleged “first liner layer,” then it would be impossible for the POSA to determine what materials must “conformally coat the contact opening” as claimed. *See id.*

56. In addition, under Complainant's interpretation, a POSA would have been unable to distinguish between a "first liner layer" and "refractory metal layer" in the case where more than two (for the '546 patent) or three (for the '167 patent) coatings of materials could qualify either alone or in combination to practice either or both of these limitations. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] These sorts of word games demonstrate the meaninglessness and indefiniteness of the “layer” terms under Complainant’s interpretations.² A POSA would have no idea how to assess infringement or non-infringement in view of the arbitrary nature of line drawing enabled by these indefinite interpretations.

59. According to Complainant, the ’546 and ’167 patents do not use the term “layer” in accordance with its plain and ordinary meaning. Instead, according to Complainant’s Infringement and Domestic Industry Contentions, the term “layer” is a meaningless placeholder for potentially co-mingled, overlapping, and/or indistinguishable areas of material, as used in the ’546 and ’167 patents. For example, the independent claims of the ’546 patent and ’167 patent refer to a “dielectric layer,” “first liner layer,” and “refractory metal layer.” *See* ’546 patent, claims 1, 8; ’167 patent, claim 1. However, the claims provide no guidance as to how to identify any layer or how to distinguish any layer from any other component or structure recited in the claims. Applying Complainant’s interpretation of the claims, a POSA would not know whether a

² I have been informed by counsel that the same claim terms are generally interpreted to have the same meanings in different but related patents, unless the claim language expressly requires otherwise.

distinguishable thickness of dielectric material should be considered a single “dielectric layer” or multiple, arbitrarily identified layers or thicknesses of the same (or different) material(s). Likewise, a POSA would not know from the claims whether two separate layers of dielectric material are a single dielectric “layer” or multiple layers.

60. Figure 1 from the common specification of the '546 and '167 patents illustrates the lack of reasonable certainty regarding the term “layer” under Complainant’s interpretation within the context of references to a “dielectric layer” (item 120). '546 patent, 2:56-57.

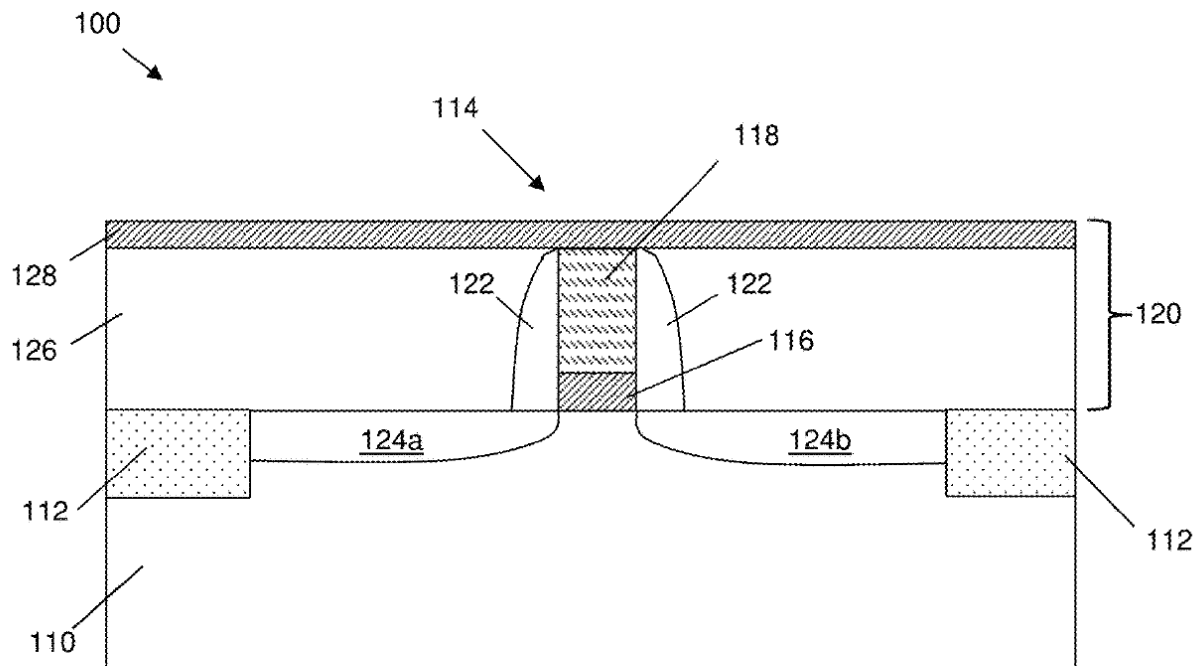


FIG. 1

The common specification states: “In one embodiment of the invention, formation of dielectric layer 120 may include deposition of a dielectric material 126 and a dielectric cap material 128.” *Id.*, 4:56-58. The specification does not explain why a POSA would consider these two dielectric “material[s]” to be a single “layer” instead of separate “layers.” In fact, the specification compounds this uncertainty by stating: “While shown as two separate layers, it is to be understood

that dielectric layer 120 is not limited to only two layers.” *Id.*, 4:58-60. “Rather, in other embodiments of the invention, dielectric layer 120 may include a single layer or more than two separate layers.” *Id.*, 4:60-62. As such, the term “layer,” as used in the patents, could mean a “single layer,” two layers, or “more than two” layers. *Id.*, 4:56-62. Applying Complainant’s departure from the plain and ordinary meaning of these terms, a POSA could not reliably identify any “layer” under this term’s meaning as interpreted by Complainant. For instance, layers within a larger layer could comprise the same material, making it difficult, if not impossible, for a POSA to distinguish between layers when attempting to assess infringement or non-infringement. Similarly, a POSA would not know whether or when to combine different, distinguishable layers of material to consider them to be a single “layer.”

61. The term “first liner layer” is similarly recited in all claims of the ’546 and ’167 patents without any guidance as to what might qualify as a “first liner layer.” Claim 1 of the ’546 patent requires “a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate.” *See* ’546 patent, claim 1; *see also id.*, claim 8 (reciting similar “first intermix region” over source and drain regions). The POSA would have been unable to assess infringement of this claim element if “first liner layer” were not interpreted to mean “a separate and defined [first liner] coating of material,” in other words, the first coating of material that is deposited to “conformally coat the contact opening.” The ’546 patent’s independent claims (1 and 8) require the recited first liner layer to “conformally coat” one or more contact openings and require “a portion of the first liner layer to diffuse” into the upper portion of substrate, source, or drain regions. *See* ’546 patent, claims 1, 8. The ’167 patent’s claims require the first liner layer to “substantially coat[] a sidewall” of a contact opening. But the claims do not recite any limitations that a POSA would interpret as

rendering the term “first liner layer” definite. For example, the claims provide no guidance as to how to identify any first liner layer or how to distinguish any first liner layer from any other component or structure recited in the claims. A POSA would not know whether a distinguishable thickness of any material should be considered a single “first liner layer” or multiple, arbitrarily identified first liner layers or thicknesses of the same (or different) material(s). Likewise, a POSA would not know from the claims whether two separate layers of first liner layer material are a single “first liner layer” or multiple layers. Layers within a larger layer could comprise the same material, making it difficult, if not impossible, for a POSA to distinguish between layers when attempting to assess infringement or non-infringement of the “first liner layer” term. Similarly, a POSA would not know whether or when to combine different, distinguishable layers of material to consider them to be a single “first liner layer.”

62. According to the patents’ common specification, “a first liner layer 132 may be deposited to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128,” as shown in Figure 3. ’546 patent, 5:57-59.

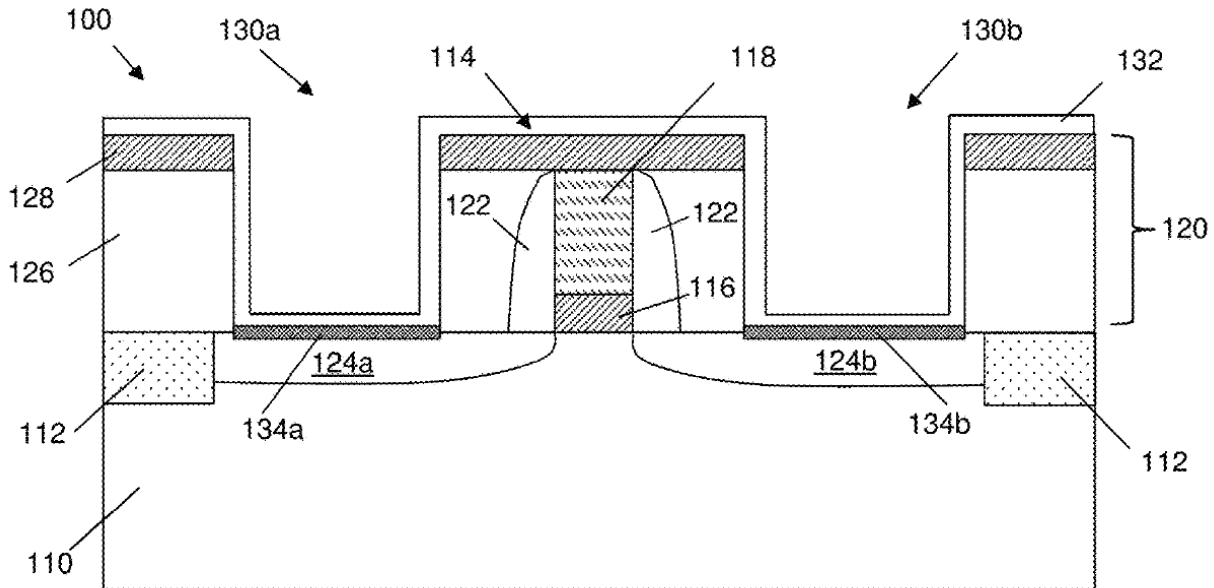


FIG. 3

“First liner layer 132 may include, but is not limited to, at least one of: NiPt, Ni, Ti, TiSi, Co, Er, Pt, and Yb.” *Id.*, 5:64-65. Again, the first liner layer could comprise one or more layers of any of these materials or any other material(s). There is no requirement for the different sublayers of a “first liner layer” to be distinguishable or comprised of different materials.

63. In another part of the specification, the patents confusingly refer to both the originally deposited “first liner layer” and the undiffused “first liner layer” as being a “first liner layer,” even though the latter does not include portions of the original first liner layer that have diffused into substrate. ’546 patent, 6:48-63. The specification states: “Referring now to FIG. 4, another step of the method according to embodiments of the invention is shown.” *Id.*, 6:48-49. Figure 4 is depicted below:

The specification states: “In this step, first liner layer 132 is removed from each contact opening 130a, 130b as well as from the upper surface of dielectric cap material 128.” *Id.*, 6:49-52. This confirms that a POSA would have been confused in view of the patents’ uncertain teachings about the “layer” terms, given that the patents refer to both a larger “first liner layer” and only a portion of this same, larger layer as a “first liner layer.” If a “first liner layer” were arbitrarily deemed to comprise multiple sublayers, a POSA would not know, from the patents’ teachings, whether all or only some of such sublayers must form part of the claimed “intermix region” in the ’167 patents’ claims or “first” or “second” “intermix region[s]” in the ’546 patents’ claims, further confirming that these terms would not have been reasonably certain to a POSA.

33

need not serve any particular function or purpose in the '546 patent's claims. The '167 patent's claims require the refractory metal layer to "substantially coat[]" whatever may be arbitrarily identified as "the first liner layer" (as discussed above). The claims also require the refractory metal layer to include a "multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal." '167 patent, claim 1. Again, this only confirms the indefiniteness of the refractory metal layer term under Complainants interpretation, because it can be arbitrarily identified as including any two or more "refractory metal" materials, which is yet another term that the specification does not use in accordance with its plain and ordinary meaning. *See supra*, n.1.

65. The claims provide no guidance as to how to identify any refractory metal layer or how to distinguish any refractory metal layer from any other component or structure recited in the claims. For example, aside from needing to separate whatever is arbitrarily identified under Complainant's interpretation as a "first liner layer" and a "metal" filler from the sublayers of the claimed "refractory metal layer" in the '167 patent, there is no function associated with the refractory metal layer that could distinguish it from any other layer of material(s). A POSA would not know whether a distinguishable thickness of any material (with respect to the '546 patent) or any two or more "refractory metal materials" (with respect to the '167 patent) should be considered a single "refractory metal layer" or multiple, arbitrarily identified refractory metal layers or thicknesses of the same (or different) material(s) under Complainant's interpretations. Likewise, a POSA would not know from the claims whether two (for the '546 patent) or three (for the '167 patent) or more separate layers of refractory metal layer material are a single "refractory metal

layer” or multiple layers. Layers within a larger layer could comprise the same material, making it difficult, if not impossible, for a POSA to distinguish between layers when attempting to assess infringement or non-infringement of the “refractory metal layer” term. Similarly, a POSA would not know whether or when to combine different, distinguishable layers of material to consider them to be all or part of a “refractory metal layer.”

66. The common specification confirms indefiniteness of the “refractory metal layer” term under Complainant’s interpretation. “As shown in FIG. 5, another aspect of the method according to embodiments of the invention is shown to include depositing a refractory metal layer 140 to conformally coat each contact opening 130a, 130b and substantially cover dielectric cap material 128 and first intermix regions 134a, 134b at the bottom portion of each contact opening 130a, 130b.” ’546 patent, 6:64-7:3.

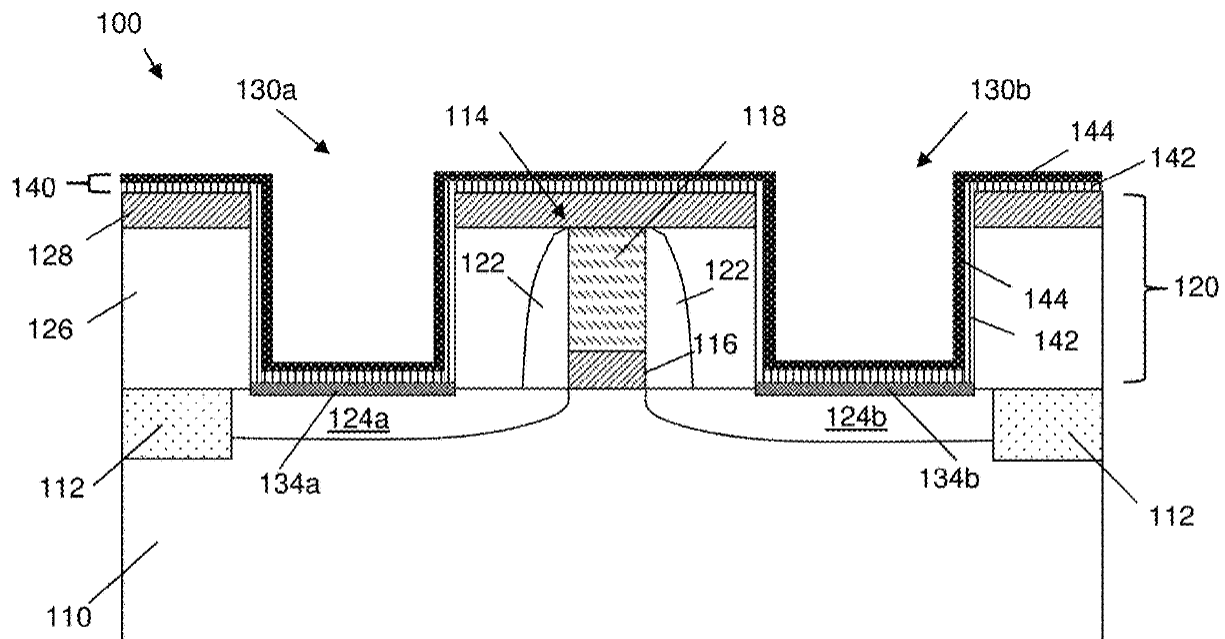


FIG. 5

According to the specification, “[r]efractory metal layer 140 may include, but is not limited to: W, Ti, Ta or Mo employed directly as a ‘metal’; a refractory metal employed as a constituent of a ‘compound’ thereof, e.g., TiN, TaN, etc.; or even as a constituent of an ‘alloy’ with another metal, e.g., titanium-tungsten.” *Id.*, 7:3-8. In other words, the refractory metal layer can be composed of any metal. “In some embodiments, as shown in FIG. 5, refractory metal layer 140 may include a bilayer of a first refractory material 142 and a second refractory material 144 such that first refractory metal material 142 substantially separates first liner layer 132 from second refractory metal layer 144.” ’546 patent, 7:8-13. “For example, in some embodiments, first refractory metal material 142 may include Ti and second refractory metal material 144 may include TiN.” *Id.*, 7:15-17. “While shown as a bilayer, it is to be understood that refractory metal layer 140 may include any number of layers of refractory metal materials without departing from aspects of the invention.” *Id.*, 7:20-23. Complainant appears to rely on this disclosure to interpret “layer” to mean “any number of layers.” Under Complainant’s interpretation, there is also no requirement that multiple sublayers in a “refractory metal layer” comprise different metals or that the sublayers be distinguishable from one another, making it impossible for the POSA to assess infringement or non-infringement.

67. For the reasons above, it is my opinion that the terms “layer,” “first liner layer,” and “refractory metal layer” lack reasonable certainty and are indefinite if not interpreted in accordance with Respondents’ proposed plain and ordinary meaning construction of “a separate and defined [first liner / refractory metal] coating of material.”

B. “intermix region”

Disputed Claim Term	Complainant’s Proposed Construction	Respondents’ Proposed Construction
“intermix region” (’546 patent, claims 1, 2, 6, 8, 9, and 13; ’167 patent, claims 1, 2, and 6)	Plain and ordinary meaning; not indefinite.	Indefinite

68. In my opinion, the asserted claims of the ’546 and ’167 patents are indefinite because the term “intermix region” fails to inform a POSA about the scope of the alleged inventions with reasonable certainty, when read in light of the patents’ specifications and prosecution histories.

69. The term “intermix region” is not a term that a POSA would have recognized to have any particular meaning in the art. For example, Respondents produced dozens of prior art references with their Invalidity Contentions related to the ’546 and ’167 patents. I ran searches in these references and did not find any uses of the term “intermix region.” Thus, a POSA would have looked to the common specification of the ’546 and ’167 patents to attempt to determine the meaning of “intermix region.” However, as summarized above in § IV, the specification puts no restrictions on what qualifies as an “intermix region.” For example, the patents do not provide any teachings or guidance as to whether an uneven surface of two interfacing materials might qualify as an “intermix region.” Alternatively, the patents do not explain if any kind of purposeful intermixing of materials is required or whether incidental or interstitial intermixing resulting from natural processes (not part of manufacturing process steps) could create an intermix region.

70. In addition, the patents provide no guidance as to whether an intermix region must have a uniform distribution of intermixed materials. For example, the “intermix region” in the ’167 patent’s claims and the “second intermix region” in claims 6 and 13 of the ’546 patent must

be composed of materials from a “first liner layer,” “semiconductor substrate,” and “refractory metal layer.” *See* ’167 patent, claim 1; ’546 patent, claims 6 and 13. But there is no explanation in the patents as to whether this second intermix region may have, for example, intermixing of the refractory metal layer with the first liner layer at the top of the alleged intermix region (with no intermixed material from the semiconductor substrate included) and intermixing of the first liner layer with material from the semiconductor substrate at the bottom of the alleged intermix region (with no intermixed material from the refractory metal layer included), or, alternatively, whether substantially uniform intermixing of all three materials is required throughout, or, alternatively, whether portions of relatively lower (or no) intermixing may be included. Similarly, for the “first intermix region” in the ’546 patent’s claims, the patents do not explain whether intermixing between the first liner layer and semiconductor substrate must be uniform or whether portions of relatively lower (or no) intermixing may be included. *See* ’546 patent, claims 1, 8.

71. According to the patents: “First liner layer 132 may be deposited such that a first intermix region 134a, 134b is formed at an interface of the bottom portion of each contact opening 130a, 130b and the upper portion of each of source 124a and drain 124b.” ’546 patent, 5:65-6:2. “First intermix regions 134a, 134b may originate from the high energy of the incoming sputtered atoms (from first liner layer 132) which are thus pushed into the Si lattice of source 124a and drain 124b.” *Id.*, 6:3-6. “Therefore, first intermix regions 134a, 134b may include a mixture of the material used for first liner layer 132 and a material used for the respective semiconductor substrate 110 thereunder.” *Id.*, 6:6-9. “In the embodiment shown in FIG. 3, first intermix regions 134a, 134b may have a composition including a material used for first liner layer 132 and materials used for the respective source and drain 124a, 124b thereunder.” *Id.*, 6:9-13. “While the Si lattice is a dense way to pack Si atoms, it has a relatively open structure due to its bonding being covalent.”

Id., 6:13-15. “The first few metal atoms, arriving with relatively high energy, can be forced into the interstitial sites of the Si lattice to create a thin region (intermix regions 134a, 134b) of mixed Si and metal on the order of 3-5 nm in thickness, for example.” *Id.*, 6:15-19. The patents provide no explanation as to whether portions of the first liner layer that **do not** penetrate semiconductor substrate may be included in any alleged “intermix region.”

72. The common specification continues: “In some cases, first intermix region 134a, 134b may have a relatively high Si content (greater than 50% Si in atomic composition).” ’546 patent, 6:19-21. As noted above, the patents put no upper limit on the amount of “Si content” or substrate material an “intermix region” may contain while still qualifying as an “intermix region.” The patents state: “In other embodiments, the compositions of first intermix regions 134a, 134b may vary depending on one or more factors including, but not limited to: the type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.).” *Id.*, 6:21-28. “In some embodiments, it may be desirable to select a target composition of first intermix regions 134a, 134b in view of one or more criteria, e.g., etch selectivity.” *Id.*, 6:28-31. “First intermix region 134a, 134b may be amorphous and a subsequent anneal may crystallize intermix region 134a, 134b into a lower resistance layer as will be described herein.” *Id.*, 6:31-34. In sum, as discussed above, the common specification puts no meaningful limits on what could qualify as an “intermix region.” It could be composed of any materials in any relative concentrations created via any method involving varying or non-varying levels of uniformity.

73. The specification further states: “First intermix regions 134a, 134b allow for formation of a very thin contact layer that is normally very difficult to control with standard

sputtering tools.” ’546 patent, 6:35-37. However, as noted above, the specification does not explain what the inventors considered to be “standard sputtering tools” and what tools were not “standard.” *Id.* The POSA would not understand how to distinguish “standard sputtering tools” from those that the specification identifies as being used to make the alleged inventions (including “ionized PVD” and “radio frequency PVD (RFPVD) or other directional energetic thin film deposition techniques such as [i]on beam assisted deposition.” *Id.*, 5:57-64. In my undergraduate class that I have been teaching at UT for 40 years, I discuss a wide variety of sputtering tools under the rubric of “standard.” These include DC and RF sputtering, reactive sputtering which involve some chemical reactions in addition to PVD, ion beam assisted deposition (IBAD), magnetron sputtering and several others. Variants of these are used in research and industry and would qualify as “standard” in my opinion.

74. The specification continues: “First intermix regions 134a, 134b are much more conformal on Si structures that are not planar.” ’546 patent, 6:37-39. But the claims of the ’546 and ’167 patents do not require any particular level of conformality on Si structures. “Additionally, upon annealing of first intermix regions 134a, 134b (which will be described further herein), very little if any metal diffusion is needed to form the low resistance contact layer which is often a source of defects in small devices.” *Id.*, 6:39-43. Thus, according to the patent, “very little if any metal diffusion is needed” to form the intermix region into a low resistance contact layer, which teaching would have further confused the POSA as to what, if any, amount of intermixing is required.

75. The specification continues: “As shown in FIG. 8, a laser anneal process may be performed such that a second intermix region 154a, 154b is formed at the interface of the bottom

portion of contacts 170a, 170b and upper portion of source 124a and drain 124b.” ’546 patent, 7:59-63.

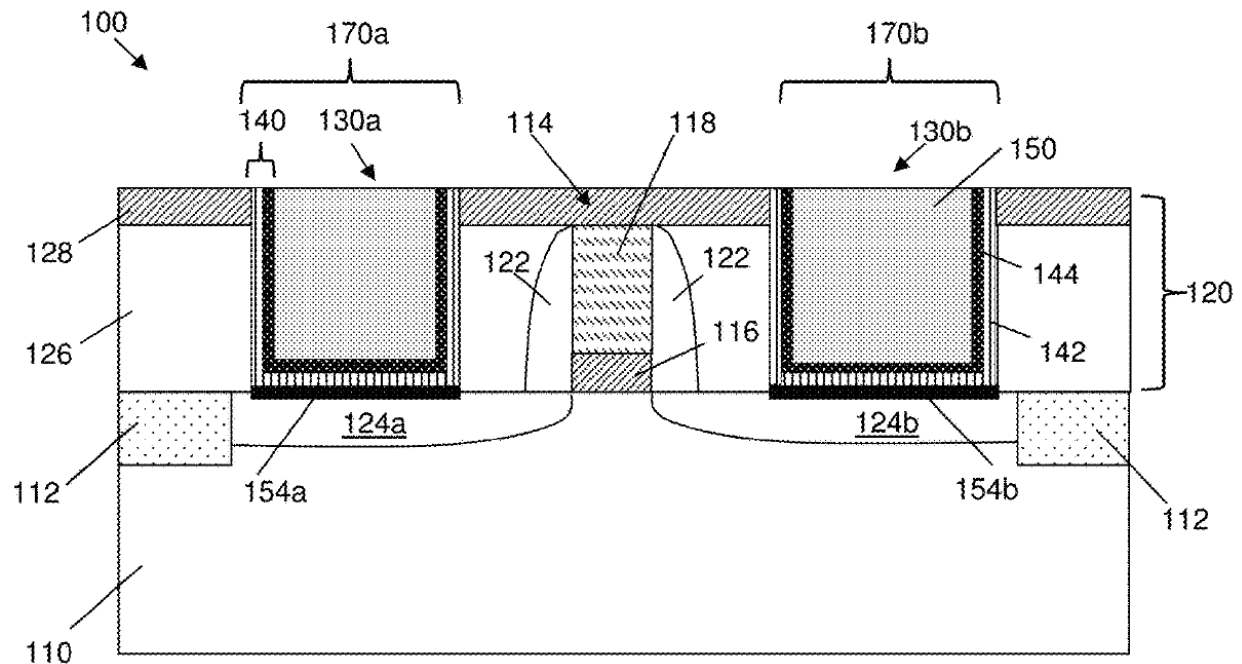


FIG. 8

“The laser anneal process may transform the composition of first intermix region 134a, 134b (FIGS. 3-7) into a second composition resulting in second intermix region 154a, 154b in place of first intermix region 134a, 134b.” *Id.*, 7:64-8:1. “That is, the laser anneal process may crystalize intermix regions 134a, 134b into a lower resistance layer, i.e. a second intermix region 154a, 154b.” *Id.*, 8:1-3. “Second intermix regions 154a, 154b may have compositions including a mixture of the materials used for source 124a and drain 124b, first liner layer 132, and refractory metal layer 140.” *Id.*, 8:3-6. “In an embodiment where Ti is used as refractory metal layer 140, second intermix region 154a, 154b may be a silicide, such as but not limited to: TiSi, NiSi, or NiPtTiSi.” *Id.*, 8:6-9. “That is, during the laser anneal process, refractory metal 140 may diffuse

into first intermix regions 134a, 134b (FIGS. 3-7) to create second intermix regions 154a, 154b.”
Id., 8:9-12.

76. As noted above, the specification includes an embodiment in which “first intermix region 134a, 134b may include TiSi and second intermix region 154a, 154b may include TiSi with a higher concentration of Ti.” ’546 patent, 9:1-4. The specification does not require the first intermix region to have a composition that is different from the second intermix region, as demonstrated by this embodiment. There is no requirement regarding the homogeneity (or lack thereof) of any “intermix region” in terms of how uniformly components of the region are distributed.

77. The term “intermix” is defined in one dictionary to mean “intermingled” or “to become mixed together.” Ex. 12, Intermix, *Merriam-Webster’s Collegiate Dictionary*, at 653 (11th ed. 2003). While this term may have had a plain and ordinary meaning consistent with this definition at the time of alleged invention, the ’546 and ’167 patents do not use this term in accordance with such plain and ordinary meaning. On the contrary, as noted above, there is no requirement for intermingling or mixing things together in the patents. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

78. Complainant's Infringement Contentions likewise demonstrate the indefiniteness of this term. For example, Complainant identifies, among other things, separate and non-intermixed titanium and nickel layers in its mapping for the "first intermix region" claim element. Ex. 1, Complainant's Infringement Contentions, Ex. A4 at 7-12. Again, if subject matter experts associated with Complainant are unable to coherently determine what is or is not an intermix region, a POSA (presumably having substantially less expertise than such individuals) would be unable to understand the scope of these terms with reasonable certainty. The term "intermix region" lacks reasonable certainty and is therefore indefinite.

79. The patents lack sufficient guidance as to what it means to be "intermixed." For example, the figure below from Ex. 15, Chow, et al., *Silicide Formation from Laser Thermal Processing of Ti/Co Bilayers* (2004), at 2 (Fig. 3), shows layers of TiO_x , a "bilayered structure" of "amorphous CoTi silicide" "on a highly textured CoTi silicide layer, and Si substrate."

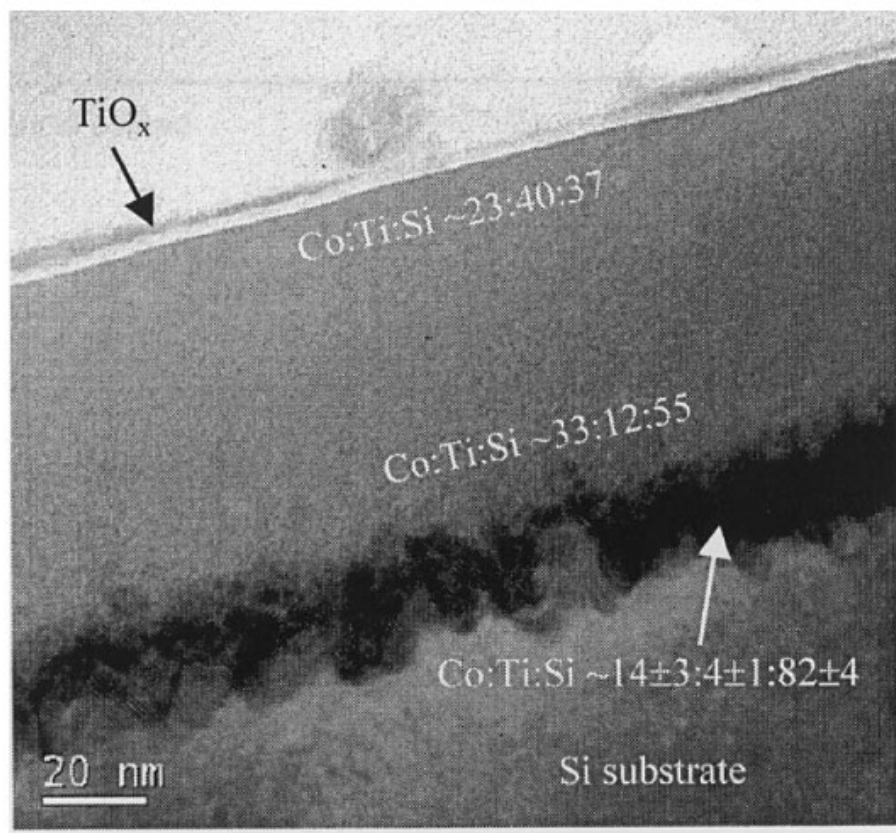


Figure 3. XTEM of the Ti/Co/Si sample after pulsed laser irradiation at 0.6 J/cm² shown in Fig. 1. A bilayered structure, *i.e.*, an amorphous CoTi silicide layer on a highly textured CoTi silicide layer, can be seen.

The interfaces between the various layers shown above are uneven. Do these uneven surfaces form part of an “intermix region” as claimed? If so, which portions of which layers should be included in the “intermix region”? If these uneven surfaces are not sufficient to form an “intermix region,” what more is required? For example, is it enough that atoms from one coating are present in another coating, or must there be more meaningful interaction between the atoms from two layers. The patents lack adequate guidance on this issue.

80. Even to the extent the patents provide guidance on what it means to be “intermixed” (they do not), the patents provide no guidance on the *amount* of “intermixing” that is required. For

example, assuming that “intermixing” includes diffusion of atoms from one coating into another coating, is there some threshold level of diffusion that is required? The patents include a statement that “very little if any metal diffusion is needed” to form the intermix region, but lack any guidance on what qualifies as “very little.” ’546 patent, 6:39-43. For example, if a single atom from one layer diffuses into another layer, is this sufficient to qualify as an “intermix region”? If a region is 98% of one material, and only 2% of a second material, is this enough to comprise an “intermix region”? One portion of the common specification notes that the “intermix region” may have around 50% mixing between two materials. *Id.*, 6:19-21. But the specification goes on to identify numerous factors that may impact the amount of intermixing. *Id.*, 6:21-28 (“type of material being deposited, the composition of source 124a and drain 124b, the energy of the atoms of the material being deposited, and/or the conditions of the deposition technique (temperature, pressure, reaction, atmosphere, etc.)”), 6:28-31 (“etch selectivity”), 6:31-34 (“amorphous” versus “crystalline”). While the common specification identifies factors that may impact the amount of intermixing, it puts no meaningful limits on what amount of mixing would or would not qualify as creating an “intermix region.”

81. The patents do not explain if any kind of purposeful intermixing of materials is required or whether incidental intermixing (which may not be intended or desired as part of manufacturing) could create an intermix region.

82. I reserve the right to supplement this declaration with new information and/or documents that may be discovered or produced in this case, or to address any new claim constructions offered by Complainant.

PUBLIC VERSION

83. I swear under penalty of perjury under the laws of the United States that the facts set forth herein are true and correct to the best of my knowledge. I am willing to testify as to the truth and accuracy of the foregoing at any hearing in the above-captioned Investigation.

Dated: August 25, 2026

Signed:



Sanjay K. Banerjee, Ph.D.